Statistical Methods 3: Continuous Random Variables and the Central Limit Theorem

Continuous Random Variables

A function $X : \Omega \to \mathbb{R}$ is said to be a random variable if and only if for all real numbers a, the set $\{\omega \in \Omega : X(\omega) \leq a\}$ is an event. The function $F(x) = P(X \leq x)$ is called the distribution function, or Cumulative Distribution Function (CDF):

$$F(x) = \int_{-\infty}^{x} f(t)\,dt \quad \text{for all } x$$

If such a function $f(t)$ exists, then it is called a density of X, probability density function (PDF).

Exponential Random Variables

Let $\lambda > 0$ be some parameter. We can define a density:

$$f(x) = \begin{cases} \lambda \exp(-\lambda x) & x \geq 0 \\ 0 & \text{else} \end{cases}$$

such that

$$F(x) = \begin{cases} 1 - \exp(-\lambda x) & x > 0 \\ 0 & \text{else} \end{cases}$$

is the distribution function (i.e. CDF). This defines an exponential random variable.

Normal Distribution

Let $\mu \in \mathbb{R}$ and $\sigma > 0$ be some parameters.

$$f(x) = \phi_{\mu,\sigma^2}(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\}$$

This is the PDF of the normal distribution (a Gaussian). When $\mu = 0$ and $\sigma = 1$, the CDF is denoted $\Phi(x)$ (AKA the standard normal distribution).

The Lognormal Distribution

Let parameters: $\mu \in \mathbb{R}$, $\sigma > 0$, $\gamma < x$ (where γ defines a log shift in x). The density function for the Log-normal Distribution is then:

$$f(x) = \frac{1}{(x-\gamma)\sigma\sqrt{2\pi}} \exp\left\{-\frac{(\ln(x-\gamma)-\mu)^2}{2\sigma^2}\right\}$$

Statistics and Limit Theorems

Suppose we have a random sample of identically distributed independent random variables

$X_{i=1,2,\ldots,n}$ drawn from some probability density function: $f(x \mid \vec{\theta})$, where $\vec{\theta} = (\theta_1, \theta_2, \ldots, \theta_k)$.

We now take the time to define the difference between:

parameter:	A number computable only from the entire population
statistic:	A number computed from a random sample

The $\vec{\theta}$s are parameters (E.g. μ and σ for the normal distribution).

Let's define some statistics:

We define $\langle X_n \rangle$ as the sample mean:

$$\bar{X} = \langle X_n \rangle \equiv \sum_{i=1}^{n} \frac{X_i}{n}.$$

We define the sample variance:

$$S^2 \equiv \frac{1}{n-1} \sum_{i}^{n} (X_i - \bar{X})^2$$

The Law of Large Numbers states that:

$$\text{as } n \to \infty, \quad \langle X_n \rangle \to \mu$$

Example

Consider n coin flips such that:

$$X_i = \begin{cases} +1 & \text{prob} = 1/2 \\ -1 & \text{prob} = 1/2 \end{cases}$$

Let $Y_n = X_1 + X_2 + X_3 + \ldots + X_n$. Some possible events and their probabilities:

- $P(Y_{10} \leq -2) = 0.38$ (moderate)
- $P(Y_{100} \leq -20) = 0.03$ (unlikely)
- $P(Y_{1000} \leq -200) = 1.36 \times 10^{-10}$ (impossible)

Central Limit Theorem

Suppose $X_1, X_2, \ldots$ is a sequence of n identically distributed independent random variables such that $E(X_i^2) < \infty$, with $\mu = E(X_i)$ and $\sigma^2 = E[(X_i - \mu)^2]$.

Then for any real x:

$$\boxed{P\left\{\sqrt{n}\left(\frac{\langle X_n \rangle - \mu}{\sigma}\right) \leq x\right\} \to \Phi(x) \quad \text{as } n \to \infty}$$